all_weeks

March 17, 2026

Week 1

[1]:
```
import pandas as pd
import requests
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

[2]:
```
# Brug af det nutidige API
api_nutidig = "https://data.sfgov.org/resource/wg3w-h783.json"

# Vi henter f.eks. de første 100.000 rækker for at få et godt overblik (det␣
↪tager lidt længere tid, men det er nødvendigt for at få et repræsentativt␣
↪datasæt)
# Note: der er mere end 100.000 rækker i det nutidige API, så vi kan overveje␣
↪at hente flere data senere, hvis det er nødvendigt
response = requests.get(f"{api_nutidig}?$limit=100000")
data = response.json()
df_now = pd.DataFrame(data)

print(f"Datasættet har {df_now.shape[0]} rækker og {df_now.shape[1]} kolonner.")
```

```
Datasættet har 100000 rækker og 36 kolonner.
```

[3]:
```
# Se de første rækker
print(df_now.head())

# Tæl de mest almindelige kategorier
top_crimes = df_now['incident_category'].value_counts().head(10)
print("\nTop 10 Kriminalitetskategorier:")
print(top_crimes)
```

```
         row_id        incident_datetime            incident_date  \
0  152278216710  2025-10-31T17:55:00.000  2025-10-31T00:00:00.000
1  148998204134  2025-06-13T12:41:00.000  2025-06-13T00:00:00.000
2  152328663010  2025-11-03T14:00:00.000  2025-11-03T00:00:00.000
3  149000206224  2025-05-21T00:00:00.000  2025-05-21T00:00:00.000
4  152326906304  2025-10-23T00:00:00.000  2025-10-23T00:00:00.000
```

```
  incident_time incident_year incident_day_of_week          report_datetime  \
0         17:55          2025               Friday  2025-10-31T17:55:00.000
1         12:41          2025               Friday  2025-06-13T12:46:00.000
2         14:00          2025               Monday  2025-11-03T14:00:00.000
3         00:00          2025            Wednesday  2025-05-21T08:15:00.000
4         00:00          2025             Thursday  2025-11-03T13:24:00.000

  incident_id incident_number cad_number  …               data_as_of  \
0     1522782       250613625  253042598  …  2025-11-04T09:37:41.000
1     1489982       250329888  251641497  …  2025-06-14T09:37:39.000
2     1523286       250269557        NaN  …  2025-11-04T09:37:41.000
3     1490002       256057940        NaN  …  2025-06-14T09:37:39.000
4     1523269       250618885  253071894  …  2025-11-04T09:37:41.000

            data_loaded_at :@computed_region_jwn9_ihcz  \
0  2025-11-05T09:59:28.000                          53
1  2025-06-15T09:53:25.000                          73
2  2025-11-05T09:59:28.000                         NaN
3  2025-06-15T09:53:25.000                         NaN
4  2025-11-05T09:59:28.000                           8

  :@computed_region_nqbw_i6c3 :@computed_region_26cr_cadq  \
0                           3                           2
1                         NaN                           2
2                         NaN                         NaN
3                         NaN                         NaN
4                         NaN                           4

  :@computed_region_qgnn_b9vv filed_online :@computed_region_h4ep_8xdi  \
0                           3          NaN                         NaN
1                           9          NaN                         NaN
2                         NaN          NaN                         NaN
3                         NaN         True                         NaN
4                           8          NaN                         NaN

  :@computed_region_n4xg_c4py :@computed_region_jg9y_a9du
0                         NaN                         NaN
1                         NaN                         NaN
2                         NaN                         NaN
3                         NaN                         NaN
4                         NaN                         NaN

[5 rows x 36 columns]

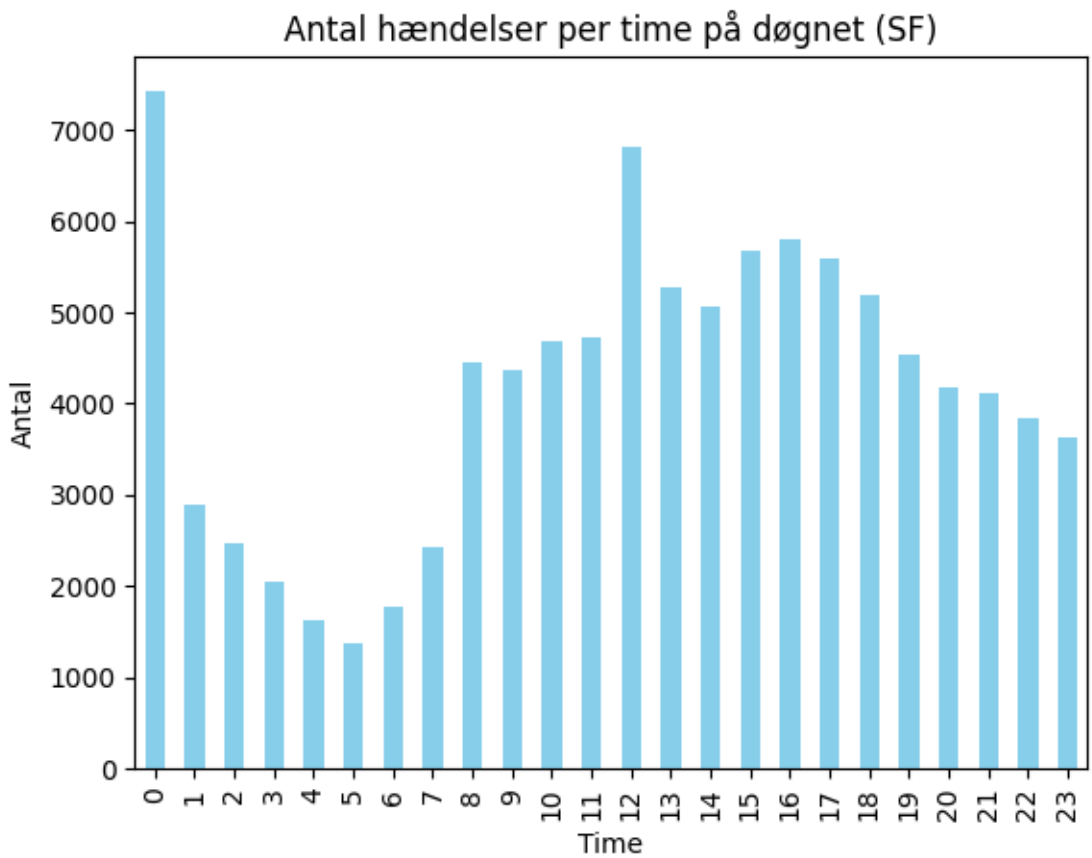
Top 10 Kriminalitetskategorier:
incident_category
Larceny Theft          21414
Assault                 7963
```

```
Drug Offense            6951
Other Miscellaneous     6782
Malicious Mischief      6328
Warrant                 5834
Burglary                4637
Motor Vehicle Theft     4265
Fraud                   4102
Non-Criminal            3970
Name: count, dtype: int64
```

[4]:
```
# Konverter tid til datetime
df_now['incident_time'] = pd.to_datetime(df_now['incident_time'], format='%H:
 ↪%M').dt.hour

# Plot antal hændelser per time
df_now['incident_time'].value_counts().sort_index().plot(kind='bar',␣
 ↪color='skyblue')
plt.title('Antal hændelser per time på døgnet (SF)')
plt.xlabel('Time')
plt.ylabel('Antal')
plt.show()

plt.savefig('week1.png')
```

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```

Week 2

[5]:
```
api_historisk = "https://data.sfgov.org/resource/tmnf-yvry.json"
res_hist = requests.get(f"{api_historisk}?$limit=100000")
df_hist = pd.DataFrame(res_hist.json())

print(f"Historisk datasæt hentet: {df_hist.shape[0]} rækker.")
```

```
Historisk datasæt hentet: 100000 rækker.
```

[6]:
```
# Eksempel på mapping (Kategori-navne kan variere lidt i API'erne)
# Vi ensretter kolonnenavne
df_hist = df_hist.rename(columns={'category': 'incident_category', 'date':␣
↪'incident_date'})

# Kombiner de to datasæt
df_combined = pd.concat([df_hist[['incident_category', 'incident_date']],
                         df_now[['incident_category', 'incident_date']]],␣
↪axis=0)
```

```
print(f"Samlet datasæt størrelse: {df_combined.shape}")
```

```
Samlet datasæt størrelse: (200000, 2)
```

Week 3

[7]:
```
# Vi kigger på Larceny/Theft i forhold til distrikter
crime_district = pd.crosstab(df_now['police_district'],␣
 ↪df_now['incident_category'])

# Normaliser for at få proportioner (betinget sandsynlighed)
crime_district_norm = crime_district.div(crime_district.sum(axis=1), axis=0)

# Vis de 5 mest almindelige forbrydelser i et bestemt distrikt (f.eks. Central)
print("Top forbrydelser i Central District (Proportioner):")
print(crime_district_norm.loc['Central'].sort_values(ascending=False).head(5))
```

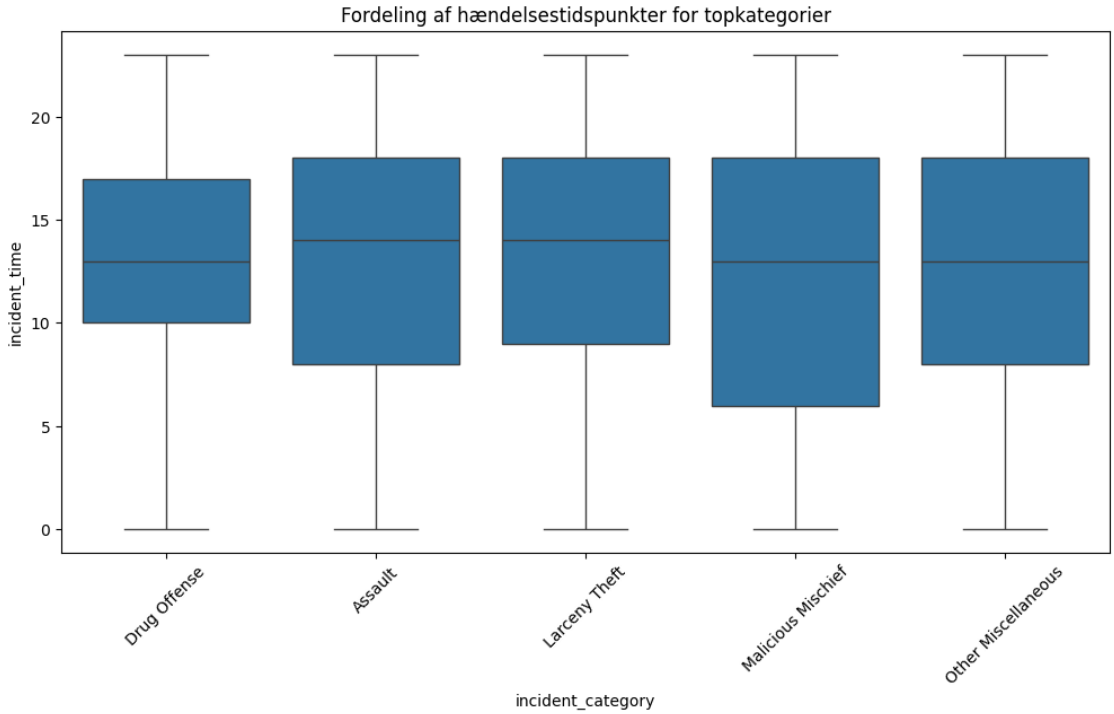
```
Top forbrydelser i Central District (Proportioner):
incident_category
Larceny Theft          0.284388
Assault                0.078015
Malicious Mischief     0.076408
Other Miscellaneous    0.066946
Lost Property          0.052932
Name: Central, dtype: float64
```

[8]:
```
# Vi fokuserer på de 5 største kategorier
import seaborn as sns

top_5_categories = df_now['incident_category'].value_counts().head(5).index
df_subset = df_now[df_now['incident_category'].isin(top_5_categories)]

plt.figure(figsize=(12, 6))
sns.boxplot(x='incident_category', y='incident_time', data=df_subset)
plt.xticks(rotation=45)
plt.title('Fordeling af hændelsestidspunkter for topkategorier')
plt.show()

plt.savefig('week3.png',)
```

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```

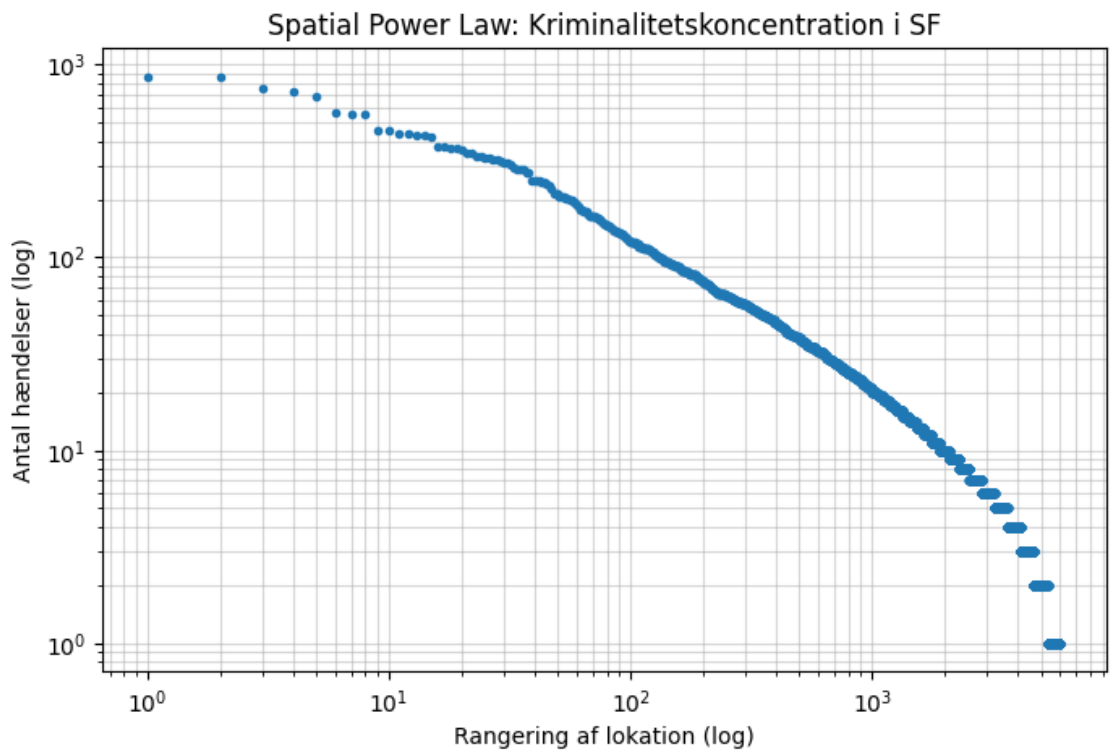
Week 4

[9]:
```
import numpy as np

# Vi tæller hændelser per lokation (lat/lon par)
location_counts = df_now.groupby(['latitude', 'longitude']).size().
 ↪sort_values(ascending=False)

# Lav log-log plot
plt.figure(figsize=(8, 5))
plt.loglog(np.arange(len(location_counts)) + 1, location_counts.values,␣
 ↪marker='.', linestyle='none')
plt.title('Spatial Power Law: Kriminalitetskoncentration i SF')
plt.xlabel('Rangering af lokation (log)')
plt.ylabel('Antal hændelser (log)')
plt.grid(True, which="both", ls="-", alpha=0.5)
plt.show()

plt.savefig('week4.png')
```




```
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```

[10]:
```
from scipy.stats import linregress

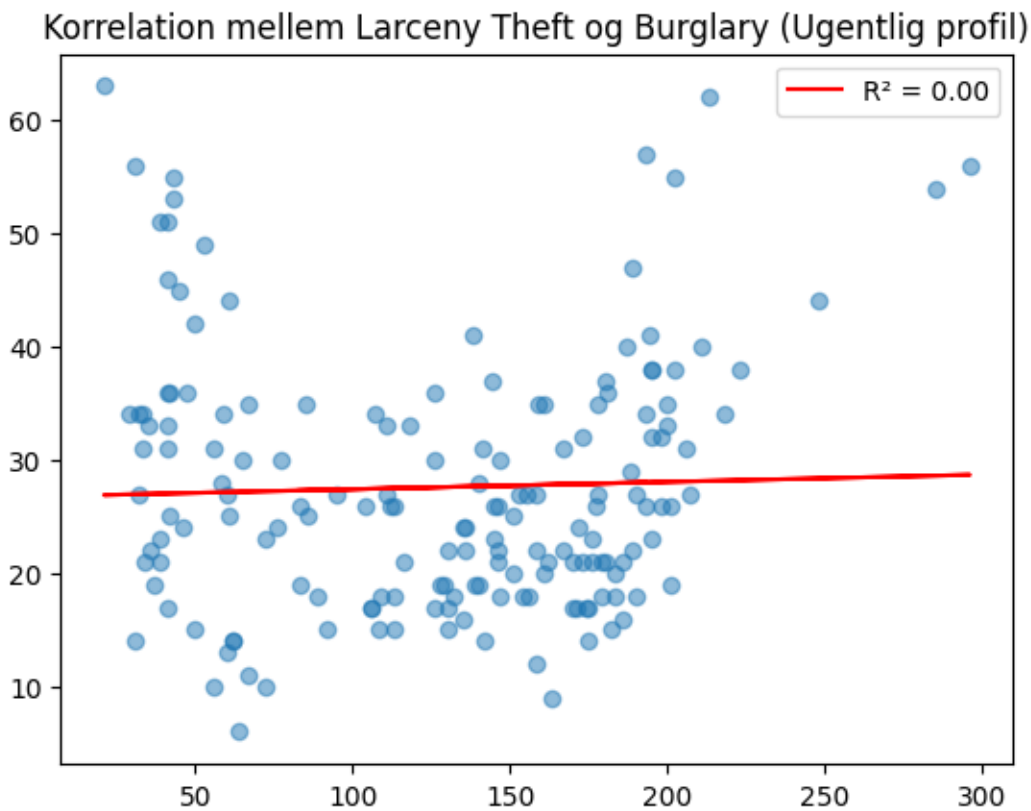
# Vi laver en tidsserie over timer i ugen (0-167) for to typer
def get_weekly_profile(crime_type):
    subset = df_now[df_now['incident_category'] == crime_type]
    # Simpel gruppering efter ugedag og time
    return subset.groupby(['incident_day_of_week', 'incident_time']).size().
 ↪values

# Eksempel med to kategorier
x = get_weekly_profile('Larceny Theft')
y = get_weekly_profile('Burglary')

# Kør regression
slope, intercept, r_value, p_value, std_err = linregress(x, y)

plt.scatter(x, y, alpha=0.5)
plt.plot(x, intercept + slope*x, 'r', label=f'R² = {r_value**2:.2f}')
plt.legend()
plt.title('Korrelation mellem Larceny Theft og Burglary (Ugentlig profil)')
plt.show()
```

```
plt.savefig('week4(1).png')
```

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```

Week 5

[11]:
```
import matplotlib.pyplot as plt
import seaborn as sns

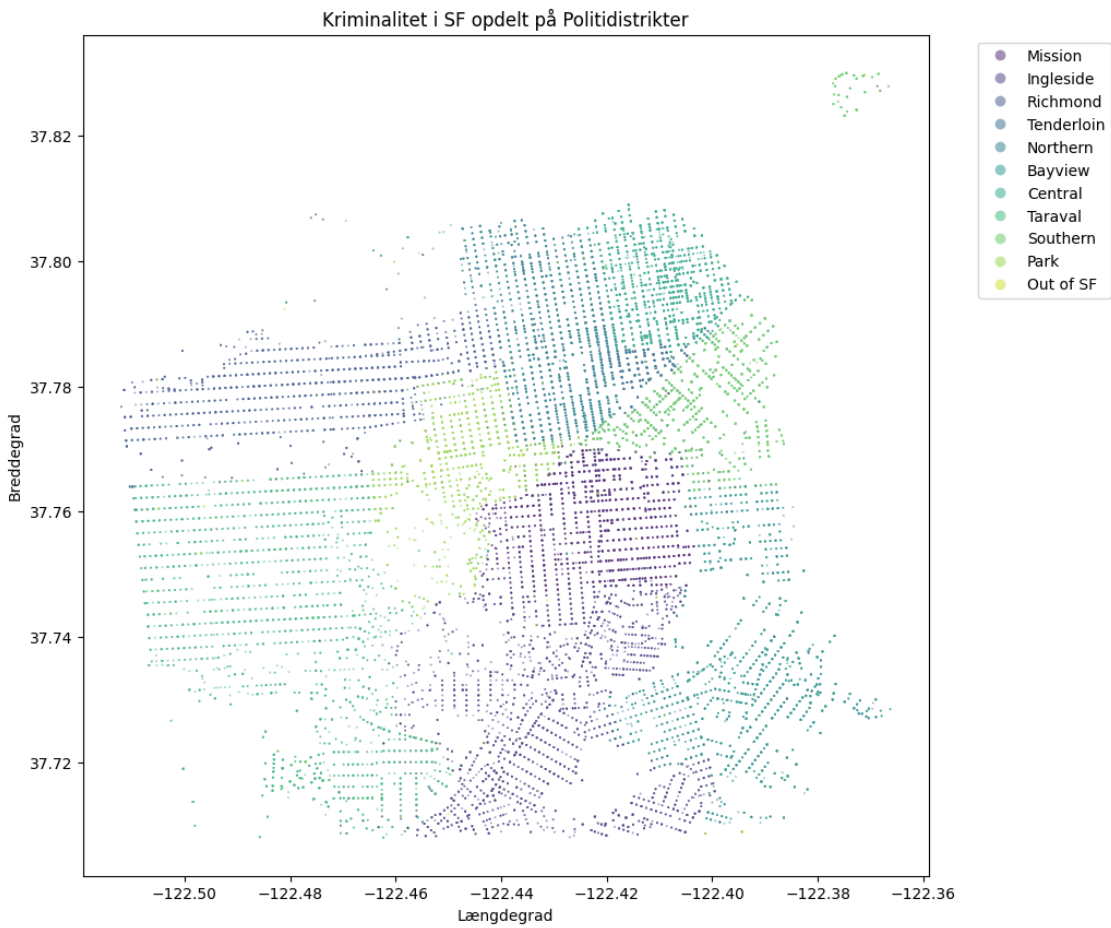
# Vi filtrerer eventuelle manglende koordinater fra
df_geo = df_now.dropna(subset=['latitude', 'longitude'])

# Sørg for at koordinater er tal
df_geo['latitude'] = df_geo['latitude'].astype(float)
df_geo['longitude'] = df_geo['longitude'].astype(float)

plt.figure(figsize=(10, 10))
# Vi bruger s=1 for at gøre punkterne små, så de ikke overlapper for meget
sns.scatterplot(data=df_geo, x='longitude', y='latitude',
                hue='police_district', s=2, alpha=0.5, palette='viridis')
```

```
plt.title('Kriminalitet i SF opdelt på Politidistrikter')
plt.xlabel('Længdegrad')
plt.ylabel('Breddegrad')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', markerscale=5)
plt.show()

plt.savefig('week5.png')
```




```
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```

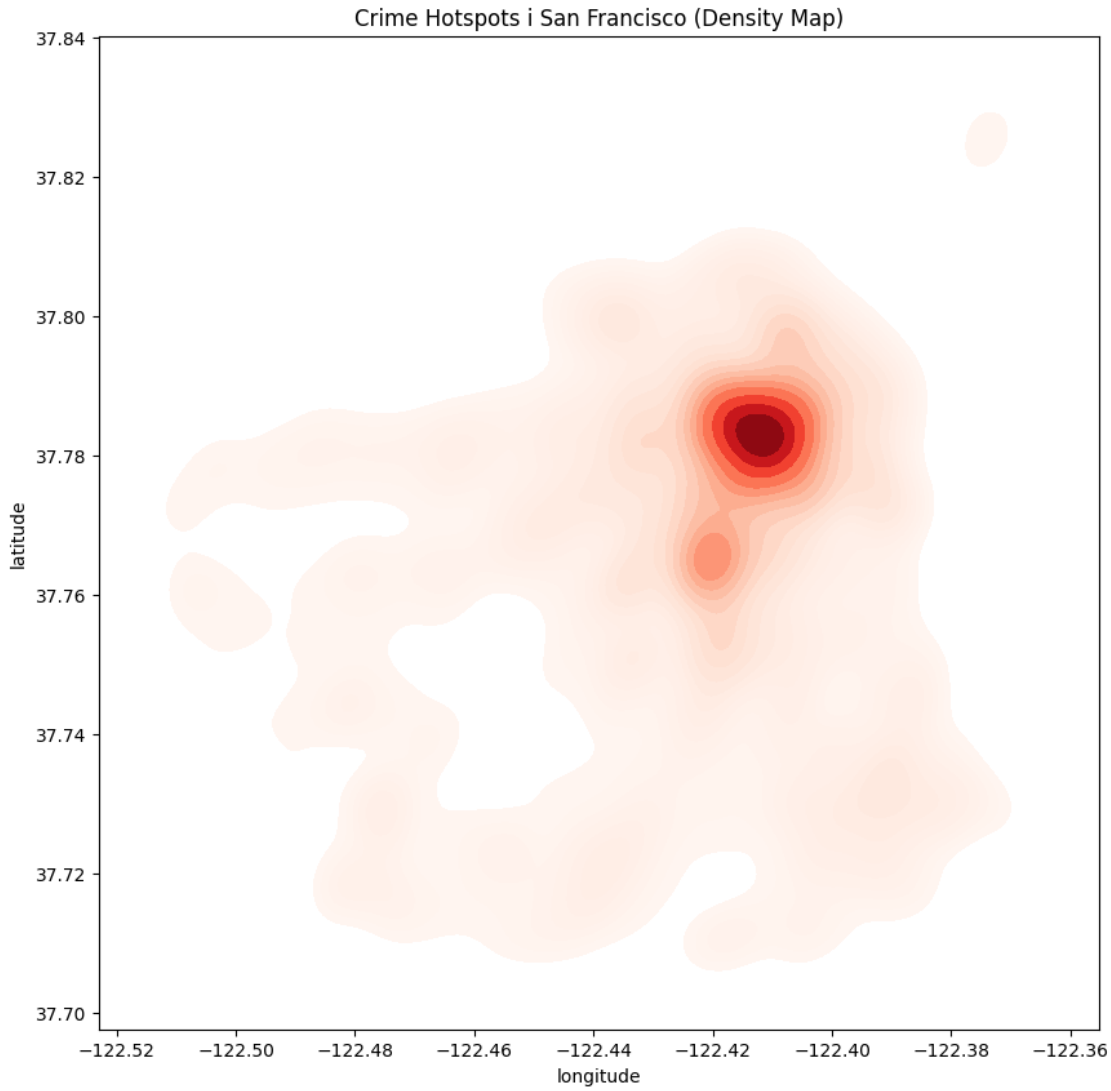
[12]:
```
plt.figure(figsize=(10, 10))

# Vi tegner tætheden af hændelser
sns.kdeplot(data=df_geo, x='longitude', y='latitude',
            fill=True, cmap='Reds', thresh=0.05, levels=20)

plt.title('Crime Hotspots i San Francisco (Density Map)')
```

```
plt.show()

plt.savefig('week5(1).png')
```




```
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```

Week 6

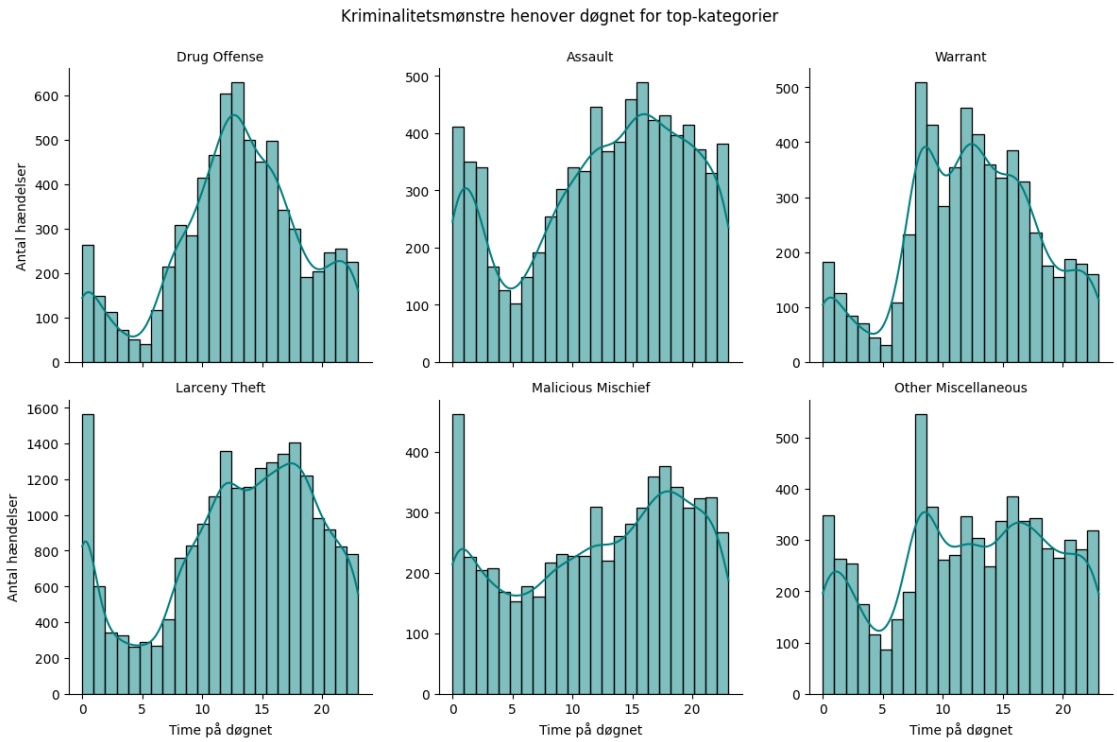
[13]:
```
# Vi vælger de top 6 mest almindelige forbrydelser
top_6 = df_now['incident_category'].value_counts().head(6).index
df_top = df_now[df_now['incident_category'].isin(top_6)]

# Opretter et gitter af grafer (Small Multiples)
```

```
g = sns.FacetGrid(df_top, col="incident_category", col_wrap=3, height=4,␣
↪sharey=False)
g.map(sns.histplot, "incident_time", bins=24, kde=True, color="teal")

g.set_axis_labels("Time på døgnet", "Antal hændelser")
g.set_titles("{col_name}")
plt.subplots_adjust(top=0.9)
g.fig.suptitle('Kriminalitetsmønstre henover døgnet for top-kategorier')
plt.show()

plt.savefig('week6.png')
```




```
<Figure size 640x480 with 0 Axes>
```